Exam-style induction problems

Cheat sheet · Day 12 · Wed 4 Nov · pack M9

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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IM FULL SOLVE (simplified circuit: RC, XM at terminals; R1 + jX + R2'/s in series)
 V1 = VL/sqrt3 (star) | VL (delta)      ns = 60f/p (p = pole PAIRS, round p=60f/n DOW
 s = (ns - n)/ns  at OPERATING speed     ws = 2*pi*ns/60,  w = 2*pi*n/60
 I2' = V1 / ((R1 + R2'/s) + jX)          IM = V1/RC - j V1/XM      I1 = I2' + IM (rec
 pf = cos(angle I1)  P_IN = 3 V1 I1 cos(phi)   Q = 3 V1 I1 sin(-phi)  (Q > 0 always)
 P_AG = 3 I2'^2 R2'/s     τ = P_AG / ws     P_OUT = τ w = (1-s) P_AG
 P_rotor,cu = s P_AG = 3 I2'^2 R2'       P_CU = 3 I2'^2 (R1+R2')   P_FE = 3 V1^2 / RC
 CHECK: P_IN - P_CU - P_FE = P_OUT       ideal eta = 1 - s (motor)
 Torque only:  τ = 3 V1^2 (R2'/s) / ( ws [ (R1+R2'/s)^2 + X^2 ] )
 GENERATOR (s<0): R2'/s, P_IN, P_AG, τ all NEGATIVE; eta_gen = |P_IN| / |τ w|
 SPEED FOR GIVEN τ_L: x = R2'/s ->  τ_L ws x^2 + (2 R1 τ_L ws - 3V1^2) x + τ_L ws (R1
   take LARGER x (small slip); quick: s ~ s_R * τ_L/τ_R;  s ~ 1/V^2 at fixed τ (low s
 Exact circuit only if asked: Vth = V1 ZM/(ZM+Z1), Zth = ZM Z1/(ZM+Z1)
 Q3 TIMING: 30 min. 3a ~8 min, 3b ~7 min, 3c ~14 min. Header block first (V1, p, ns,
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